

My Week 4 Association Rules Report

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I rebuilt the Week 4 example around the lecture's Online Retail assignment. My project now converts invoices into baskets, mines frequent itemsets with Apriori, and lets me inspect directional rules with their underlying counts. The interactive page and this report are in [my Week 4 project in the fork](#) (implementation commit **7bb475b**).

Project link: <https://github.com/PetrVorozhtsov/RecSys-LLMs/tree/main/week4>

What I used

Dataset	My preparation
UCI Online Retail; 541,909 original rows; December 2010 to December 2011	530,104 retained rows; 19,960 invoices; 3,922 distinct StockCodes
InvoiceNo = transaction; StockCode = item; Description = label	I collapsed repeated StockCodes within each invoice to binary presence.

I excluded C-prefixed cancellations, nonpositive quantity or price, and rows missing a product code, description, or date. The recorded exclusions are 9,288 cancellation/missing-invoice rows, 1,454 missing-product/name rows, 474 nonpositive-quantity rows, and 589 nonpositive-price rows. I kept one-product invoices in the denominator.

How I mined and filtered rules

I generated Apriori candidates from frequent subsets and counted basket membership exactly. At my default 1% support threshold, a rule needs at least 200 joint invoices. I chose 30% minimum confidence to require B in a substantial share of A baskets, then kept rules with lift above 1 for further analysis. These are practical starting thresholds, not universal constants.

$$\text{support} = nAB / N; \text{confidence}(A \rightarrow B) = nAB / nA; \text{lift} = \text{confidence} / (nB / N)$$

My default run found **2,026 frequent itemsets** and **2,030 retained rules**. The rejected-candidates view shows rules before the confidence/lift filter; that lets me explain why a plausible-looking association should not be used.

Key code changes

I added a reproducible UCI workbook converter, a browser/Node Apriori module, six unit tests, a measured threshold and time-split evaluation script, and an interactive interface with rule search, counts, formulas, and five live self-checks. I removed the unrelated MovieLens Two-Tower files from this Week 4 folder.

Source: Chen, D. (2015), *Online Retail*, UCI Machine Learning Repository, doi:[10.24432/C5BW33](https://doi.org/10.24432/C5BW33); CC BY 4.0. Original workbook SHA-256: 43465a06f2ccf7c8b5bd2892bc7defb52f97487934fe93b16ae4c3936424676d.

To reproduce my results, I serve the week4 folder over HTTP, then run **node apriori.test.js** and **node evaluate.js**. The committed basket JSON and evaluation JSON hold the exact counts and source checksums.

A rule I would investigate

I selected **JUMBO BAG PINK POLKADOT (22386) -> JUMBO BAG RED RETROSPOT (85099B)**. These are different bag designs that could plausibly be shown together in a cross-sell placement. This is a candidate for an experiment, not a proven sale.

01 / CONFIGURE

Find meaningful rules

Set the minimum evidence required before a rule is considered.

Minimum support

1.0%

0.5%10%

A rule's products must appear together in this share of invoices.

Minimum confidence

30.0%

5%100%

Among invoices with A, this share must also contain B.

Try a scenario

Balanced

Explore rare

High evidence

Run rules

Run tests

Opening a local file? Run `python -m http.server 8000` in the project folder and visit `localhost:8000/week4/`.

02 / EXPLORE

Frequent itemsets

2,026

at current support

Retained rules

2,030

confidence met · lift > 1

Rejected candidates

1,784

inspect weak signals

Discovered rules

1.0% support · 30.0% confidence

Retained2,030

Rejected1,784

22386 → 850998

X

JUMBO BAG PINK POLKADOT → JUMBO BAG RED RETROSPOT

StockCode 22386 → 850998

4.1%

67.7%

6.47x

support

confidence

lift

Showing 1 of 1 retained candidates. Select a row for the full calculation.

03 / INSPECT

Selected rule

JUMBO BAG PINK POLKADOT → JUMBO BAG RED RETROSPOT

StockCodes: 22386 → 850998

N · all invoices

19,960

nA · contain A

1,218

nB · contain B

2,089

nAB · contain both

825

Support

4.1%

nAB / N = 825 / 19,960

Confidence

67.7%

nAB / nA = 825 / 1,218

Lift

6.47x

confidence / (nB / N)

B occurs in 10.5% of all baskets but in 67.7% of baskets containing A. The positive lift is a discovery signal, not causal proof.

My live page at 1% support and 30% confidence. I searched for the exact directed StockCode pair and opened its counts.

N (invoices)	nA	nB	nAB	Support	Confidence	Lift
19,960	1,218	2,089	825	4.13%	67.73%	6.47x

I observed both bags in **825** invoices. The red bag's overall rate is 10.47%, but its rate among invoices with the pink bag is 67.73%; lift is 6.47. This clears my support, confidence, and lift filters.

Direction matters: for red bag -> pink bag, confidence is 39.49%, while support and lift stay the same. The antecedent has a different basket count in the reverse rule.

A candidate I would reject

I also inspected **REGENCY CAKESTAND 3 TIER (22423)** -> **JUMBO BAG RED RETROSPOT (85099B)**. It reaches the 1% joint-support floor and has lift above 1, but it is a **rejected candidate** because its confidence is below my 30% minimum.

01 / CONFIGURE

Find meaningful rules

Set the minimum evidence required before a rule is considered.

Minimum support

1.0%

0.5%10%

Minimum confidence

30.0%

5%100%

Try a scenario

BalancedExplore rareHigh evidence

Run rules

Run tests

Opening a local file? Run `python -m http.server 8080` in the project folder and visit `localhost:8080/week4/`.

WHY THESE FILTERS?

Evidence before action

At 1.0% support (at least 200 of 19,960 invoices), a rule needs enough joint transactions to be credible. At 30.0% confidence, B must occur in at least that share of baskets containing A. Lower thresholds reveal rarer patterns but add noise and computation; higher thresholds require stronger evidence. Only lift above 1 is retained for further analysis.

Rules meeting the selected thresholds with lift > 1 are retained. Rejected candidates remain available for comparison.

Frequent itemsets

2,026

at current support

Retained rules

2,030

confidence met · lift > 1

Rejected candidates

1,784

inspect weak signals

02 / EXPLORE

Discovered rules

1.0% support · 30.0% confidence

Retained 2,030Rejected 1,784

22423 → 850998

REGENCY CAKESTAND 3 TIER → JUMBO BAG RED RETROSPOT

Below confidence threshold

1.4% support14.3% confidence1.37x lift

Showing 1 of 1 rejected candidates. Select a row for the full calculation.

03 / INSPECT

Selected rule

Rejected candidate

REGENCY CAKESTAND 3 TIER → JUMBO BAG RED RETROSPOT

StockCodes: 22423 → 850998

N · all invoices19,960

nA · contain A1,988

nB · contain B2,089

nAB · contain both285

Support1.4%
nAB / N = 285 / 19,960

Confidence14.3%
nAB / nA = 285 / 1,988

Lift1.37x
confidence / (nB / N)

Below confidence threshold. This association should not be promoted as evidence for a cross-sell without further investigation.

My rejected-candidates tab at the same 1% / 30% settings. The view keeps the unfiltered candidate visible for comparison.

N (invoices)	nA	nB	nAB	Support	Confidence	Lift
19,960	1,988	2,089	285	1.43%	14.34%	1.37x

Only 285 of 1,988 cakestand invoices also contain the red bag: 14.34% confidence. That is only 3.87% above the bag's 10.47% overall basket rate, despite lift 1.37. I would not prioritize a bundle or promotion from this association.

Its lift above 1 does not override the confidence threshold, and no association alone establishes that displaying the bag causes an additional purchase.

What changed when I changed the inputs

01 / CONFIGURE

Find meaningful rules

Set the minimum evidence required before a rule is considered.

Minimum support

2.0%

0.5%

10%

A rule's products must appear together in this share of invoices.

Minimum confidence

50.0%

5%

100%

Among invoices with A, this share must also contain B.

Try a scenario

BalancedExplore rareHigh evidence

Run rules

Run tests

Opening a local file? Run `python -m http.server 8080` in the project folder and visit `localhost:8080/week4/`.

WHY THESE FILTERS?

Evidence before action

At 2.0% support (at least 400 of 19,960 invoices), a rule needs enough joint transactions to be credible. At 50.0% confidence, B must occur in at least that share of baskets containing A. Lower thresholds reveal rarer patterns but add noise and computation; higher thresholds require stronger evidence. Only lift above 1 is retained for further analysis.

1 Rules meeting the selected thresholds with lift > 1 are retained. Rejected candidates remain available for comparison.

Frequent itemsets

385

at current support

Retained rules

62

confidence met · lift > 1

Rejected candidates

128

inspect weak signals

02 / EXPLORE

Discovered rules

2.0% support · 50.0% confidence

Retained 62Rejected 128

22386 → 850998

JUMBO BAG PINK POLKADOT → JUMBO BAG RED RETROSPOT

StockCode 22386 → 850998

4.1% support67.7% confidence6.47× lift

Showing 1 of 1 retained candidates. Select a row for the full calculation.

03 / INSPECT

Selected rule

Retained rule

JUMBO BAG PINK POLKADOT → JUMBO BAG RED RETROSPOT

StockCodes: 22386 → 850998

N · all invoices19,960

nA · contain A1,218

nB · contain B2,089

nAB · contain both825

Support4.1%
nAB / N = 825 / 19,960

Confidence67.7%
nAB / nA = 825 / 1,218

Lift6.47×
confidence / (nB / N)

B occurs in 10.5% of all baskets but in 67.7% of baskets containing A. The positive lift is a discovery signal, not causal proof.

My page with the High evidence preset: 2% minimum support and 50% minimum confidence. The bag rule still passes.

Setting	Support	Confidence	Frequent sets	Retained rules
Explore rare	0.50%	20.00%	18,930	67,498
Balanced	1.00%	30.00%	2,026	2,030
Higher support	2.00%	30.00%	385	164
Higher confidence	1.00%	50.00%	2,026	995
High evidence	2.00%	50.00%	385	62

I can see the trade-off: the 0.5% / 20% setting surfaces many rare rules, while 2% / 50% leaves only 62. Stricter thresholds reduce the review workload but may hide useful niche products.

My exploratory later-period check

I split at 2011-09-01: 12,495 earlier and 7,465 later invoices. For the bag pair, confidence was 67.92% earlier and 67.31% later; lift was 6.14 and 7.11. The rejected rule stayed below 30% confidence (13.84% and 15.54%). I chose these examples using all periods, so this is a stability check rather than a blind holdout evaluation.

What I would test next

I would randomize eligible shoppers into a red-bag suggestion near the pink bag and a control with the usual page. I would compare incremental attach rate and gross margin per exposed shopper, while monitoring total conversion. I would make a sales-impact claim only after that controlled comparison.